SQL Exercise Aggregate Functions & Grouping

Database: employees_db

Id	First_Name	Last_Name	depart	Salary	hire_date	City
1	John	Doe	IT	55000	2018/06/15	NY
2	Jane	Smith	HR	48000	2019/07/20	Chicago
3	Mike	Johnson	Finance	60000	2017/04/30	LA
4	Sarah	Brown	IT	53000	2021/03/25	NY
5	David	White	Marketing	52000	2016/04/10	San Francisco
6	Emily	Davis	IT	62000	2015/02/14	Chicago
7	Robert	Wilson	Finance	59000	2019/10/01	Houston
8	Jessica	Moore	HR	51000	2018/05/22	LA
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago
10	Laura	Hall	IT	50000	2020/08/10	San Francisco

Questions

Count() Function

Write a SQL query to find the total number of employees in the company

```
SELECT COUNT(*) AS total_employees
FROM employees;
```

total_employees
10

SUM() Function

Write a SQL query to find the total Salary paid to all employees in the IT department

```
SELECT SUM(Salary) AS total_it_Salary
FROM employees
WHERE department = 'IT';
```

total_it_Salary
220000

AVG() Function

Write a SQL query to calculate the Average Salary of employees in the HR department

```
SELECT AVG(Salary) AS avg_hr_Salary
FROM employees
WHERE department = 'HR';
```

~~avg_hr_Sary~~ avg_hr_Salary
49500

MIN() and MAX() Functions

Write a SQL query to find the highest and lowest Salary in the company

```
SELECT MAX(Salary) AS highest_Salary,
       MIN(Salary) AS Lowest_Salary
FROM employees;
```

Highest_Salary	Lowest_Salary
62000	48000

Group By Statement

Write a SQL query to group employees by department and display the total Salary paid in each department

```
SELECT department
       SUM (Salary) AS total_Salary
FROM employees
GROUP BY department;
```

department	total_Salary
IT	220000
HR	9900
Finance	119000
Marketing	105000

Group By and COUNT()

Write a SQL query to count how many employees work in each city

```
SELECT City, COUNT(*) AS employees_Count
FROM employees
GROUP BY City;
```

City	Employee_Count
NY	2
Chicago	3
LA	2
San Francisco	2
Houston	1

GROUP BY and ORDER BY

Write a SQL query to group employees by department calculate the average Salary in each department, and order the results in descending order of average Salary

```
SELECT department, AVG (Salary)
       AS avg_Salary
FROM employees
GROUP BY department
ORDER BY avg_Salary DESC;
```

department	avg_Salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

Having Clause

Write a SQL query to find departments where the total Salary paid exceeds 100,000

```
SELECT department, SUM (Salary) AS
       total_Salary
FROM employees
GROUP BY department
HAVING SUM (Salary) > 100000;
```

department	total_Salary
IT	220000
Finance	119000
Marketing	105000

Combining GROUP BY, HAVING and ORDER BY

Write a SQL query to list cities where more than one employees works, Ordered by the number of Employees in descending order

City	employee_Count
Chicago	3
NY	2
LA	2
San Fransucisco	2

```
SELECT City,
       COUNT(*) AS employee_
       count
FROM employees
GROUP BY City
HAVING COUNT(*) > 1
GROUP BY employee_Count
       DESC;
```

Combining Aggregate Functions

Write a SQL Query to find the department with the highest average Salary

department	Avg-Salary
Finance	59500

```
SELECT department, AVG(Salary) AS
       avg-Salary
FROM   employees
GROUP BY department
ORDER BY avg-Salary DESC
LIMIT 1;
```